

Digital Signal Processing

A discrete signal is an infinite sequence of numbers $\{y_k\}$ where k is an integer

TABLE I Examples of Signals

Signals			
Name	Symbol	Vector	Formal Description
delta	δ	$(\dots, 0, 0, 0, 1, 0, 0, 0, \dots)$	$\{d_k\}$, where $d_k = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases}$
unit step	u	$(\dots, 0, 0, 0, 1, 1, 1, 1, \dots)$	$\{u_k\}$, where $u_k = \begin{cases} 1 & \text{if } k \geq 0 \\ 0 & \text{if } k < 0 \end{cases}$
constant	χ	$(\dots, 1, 1, 1, 1, 1, 1, 1, \dots)$	$\{c_k\}$, where $c_k = 1$

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A periodic signal is signal for which there exist a positive integer q such that $\{ p_k = p_{k+q} \}$

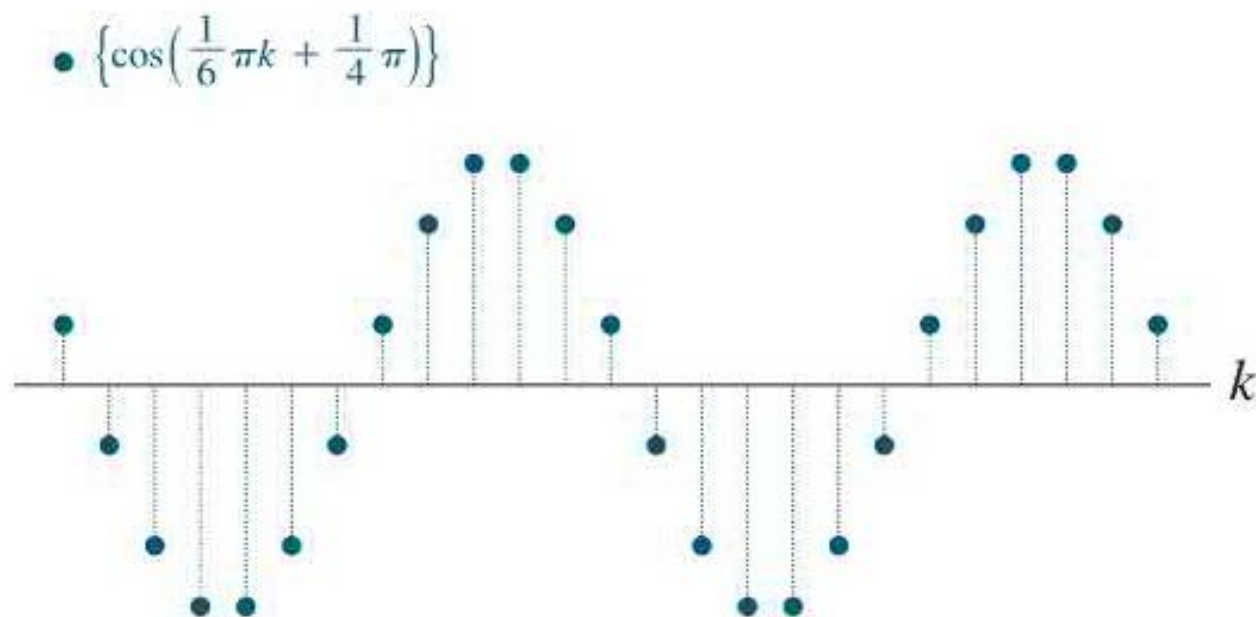


FIGURE 1

Linear Time Invariant Transformations

Linear Time Invariant (LTI) Transformations

A transformation $T : \mathbb{S} \rightarrow \mathbb{S}$ is **linear time invariant** provided

- (i) $T(\{x_k + y_k\}) = T(\{x_k\}) + T(\{y_k\})$ for all signals $\{x_k\}$ and $\{y_k\}$;
- (ii) $T(c\{x_k\}) = cT(\{x_k\})$ for all scalars c and all signals $\{x_k\}$;
- (iii) If $T(\{x_k\}) = \{y_k\}$, then $T(\{x_{k+q}\}) = \{y_{k+q}\}$ for all integers q and all signals $\{x_k\}$.

TABLE 2 Applying a Shift Signal

$\vdots$	$\vdots$	$\vdots$
$S^{-2}(\delta)$	$(\dots, 1, 0, 0, 0, \dots)$	$\{w_k\}$, where $w_k = \begin{cases} 1 & \text{if } k = -2 \\ 0 & \text{if } k \neq -2 \end{cases}$
$S^{-1}(\delta)$	$(\dots, 0, 1, 0, 0, \dots)$	$\{x_k\}$, where $x_k = \begin{cases} 1 & \text{if } k = -1 \\ 0 & \text{if } k \neq -1 \end{cases}$
δ	$(\dots, 0, 0, 1, 0, \dots)$	$\{d_k\}$, where $d_k = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{if } k \neq 0 \end{cases}$
$S^1(\delta)$	$(\dots, 0, 0, 0, 1, \dots)$	$\{y_k\}$, where $y_k = \begin{cases} 1 & \text{if } k = 1 \\ 0 & \text{if } k \neq 1 \end{cases}$
$S^2(\delta)$	$(\dots, 0, 0, 0, 0, 1, \dots)$	$\{z_k\}$, where $z_k = \begin{cases} 1 & \text{if } k = 2 \\ 0 & \text{if } k \neq 2 \end{cases}$
$\vdots$	$\uparrow$ $k = 0$	$\vdots$

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EXAMPLE 2 For any positive integer m , the **moving average** LTI transformation with time period m is given by

$$M_m(\{x_k\}) = \{y_k\} \text{ where } y_k = \frac{1}{m} \sum_{k} x_k$$

Figure 3 illustrates how M_3 smooths out a signal. Section 4.2, Figure 3 illustrates the smoothing that occurred when M_2 was applied to the same data. As m is increased, applying M_m smooths the signal even more.

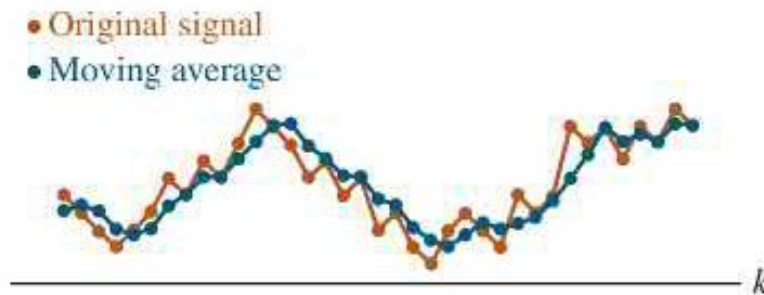
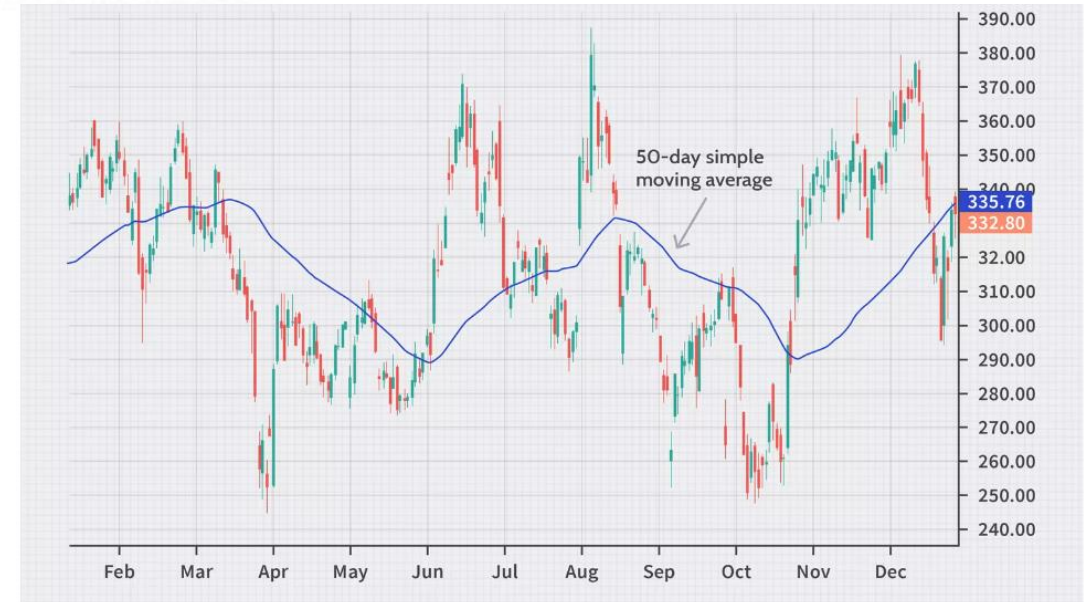


FIGURE 3



EXAMPLE 4 Using the basis $\mathcal{B}_2 = \{\delta, S(\delta), S^2(\delta)\}$ for $\mathbb{S}_2$, represent the signal $\{y_k\}$, where

$$y_k = \begin{cases} 0 & \text{if } k < 0 \text{ or } k > 3 \\ 2 & \text{if } k = 0 \\ 3 & \text{if } k = 1 \\ -1 & \text{if } k = 2 \end{cases}$$

as a vector in $\mathbb{R}^3$.

SOLUTION First write $\{y_k\}$ as a linear combination of the basis vectors in $\mathcal{B}_2$.

$$\{y_k\} = 2\delta + 3S(\delta) + (-1)S^2(\delta)$$

The coefficients of this linear combination are precisely the entries in the coordinate

vector. Thus $[\{y_k\}]_{\mathcal{B}_2} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$



Linear Difference Equations

Given scalars $a_0, \dots, a_n$, with a_0 and a_n nonzero, and given a signal $\{z_k\}$, the equation

$$a_0 y_{k+n} + a_1 y_{k+n-1} + \dots + a_{n-1} y_{k+1} + a_n y_k = z_k \quad \text{for all } k \quad (3)$$

is called a **linear difference equation** (or **linear recurrence relation**) of order n . For simplicity, a_0 is often taken equal to 1. If $\{z_k\}$ is the zero sequence, the equation is **homogeneous**; otherwise, the equation is **nonhomogeneous**.

EXAMPLE 3 Solutions of a homogeneous difference equation often have the form $\{y_k\} = \{r^k\}$ for some r . Find some solutions of the equation

$$y_{k+3} - 2y_{k+2} - 5y_{k+1} + 6y_k = 0 \quad \text{for all } k \quad (4)$$

SOLUTION Substitute r^k for y_k in the equation and factor the left side:

$$r^{k+3} - 2r^{k+2} - 5r^{k+1} + 6r^k = 0 \quad (5)$$

$$r^k(r^3 - 2r^2 - 5r + 6) = 0$$

$$r^k(r-1)(r+2)(r-3) = 0 \quad (6)$$

Since (5) is equivalent to (6), $\{r^k\}$ satisfies the difference equation (4) if and only if r satisfies (6). Thus $\{1^k\}$, $\{(-2)^k\}$, and $\{3^k\}$ are all solutions of (4). For instance, to verify that $\{3^k\}$ is a solution of (4), compute

$$\begin{aligned} & 3^{k+3} - 2 \cdot 3^{k+2} - 5 \cdot 3^{k+1} + 6 \cdot 3^k \\ &= 3^k(27 - 18 - 15 + 6) = 0 \quad \text{for all } k \end{aligned}$$



THEOREM 19

If $a_n \neq 0$ and if $\{z_k\}$ is given, the equation

$$y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = z_k \quad \text{for all } k \quad (7)$$

has a unique solution whenever $y_0, \dots, y_{n-1}$ are specified.

THEOREM 20

The set H of all solutions of the n th-order homogeneous linear difference equation

$$y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = 0 \quad \text{for all } k \quad (10)$$

is an n -dimensional vector space.

Reduction to Systems of First-Order Equations

In general, the equation

$$y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = 0 \quad \text{for all } k$$

can be rewritten as $\mathbf{x}_{k+1} = A\mathbf{x}_k$ for all k , where

$$\mathbf{x}_k = \begin{bmatrix} y_k \\ y_{k+1} \\ \vdots \\ y_{k+n-1} \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_n & -a_{n-1} & -a_{n-2} & \cdots & -a_1 \end{bmatrix}$$

EXAMPLE 6 Write the following difference equation as a first-order system:

$$y_{k+3} - 2y_{k+2} - 5y_{k+1} + 6y_k = 0 \quad \text{for all } k$$

SOLUTION For each k , set

$$\mathbf{x}_k = \begin{bmatrix} y_k \\ y_{k+1} \\ y_{k+2} \end{bmatrix}$$

$$\mathbf{x}_{k+1} = A\mathbf{x}_k \quad \text{for all } k, \quad \text{where} \quad A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & 5 & 2 \end{bmatrix} \quad \blacksquare$$